Assignment 2: Use case realizations

Due Sunday, October 25, 23:59 on Brightspace

This assignment is worth 5% of the final grade. It is submitted in teams of 2, with one submission per team. Identify all authors on the cover page of the submission by giving their names and student numbers. The submission should also include a section titled **Work distribution** that states what each member contributed.

Any submission of which you are not the author is a violation of academic integrity. This includes solutions provided by any third party, including those generated by artificial intelligence tools.

All diagrams must be made with a modelling tool or a diagramming tool such as Mermaid, draw.io, or Excalidraw.

Background

This assignment continues assignment 1 on the Airport Employee Management System (AEMS). The use cases *Register Department Employee* and *Select Work Shifts* are the same as in assignment 1.

Use case: Register Department Employee	
Primary actor	Airport Manager
Precondition	Airport Manager is authenticated
Steps	1. The Airport Manager selects to register an Employee. 2. The AEMS asks for the employee registration information. 3. The Airport Manager submits the employee registration information. 4. The AEMS registers the employee and displays the id number of the new employee. 5. The AEMS presents the list of departments. 6. The Airport Manager selects the department in which the new employee is assigned. 7. The AEMS assigns the employee to the selected department and notifies the department's manager.
Postcondition	A new employee was registered in a department and the department's manager was notified

Use case: Select Work Shifts	
Primary actor	Employee
Precondition	Employee is authenticated
Steps	1. The Employee asks to see the list of upcoming shifts in their department. 2. The AEMS displays the list of shifts.

	3. The Employee selects the shifts for which they are available and submits. 4. The AEMS records the employee's selection.
Postcondition	The preferred shifts of the employee were recorded

Domain model

Figure 1 contains domain models relevant to two use cases, separated by modules corresponding to relevant subdomains of AEMS

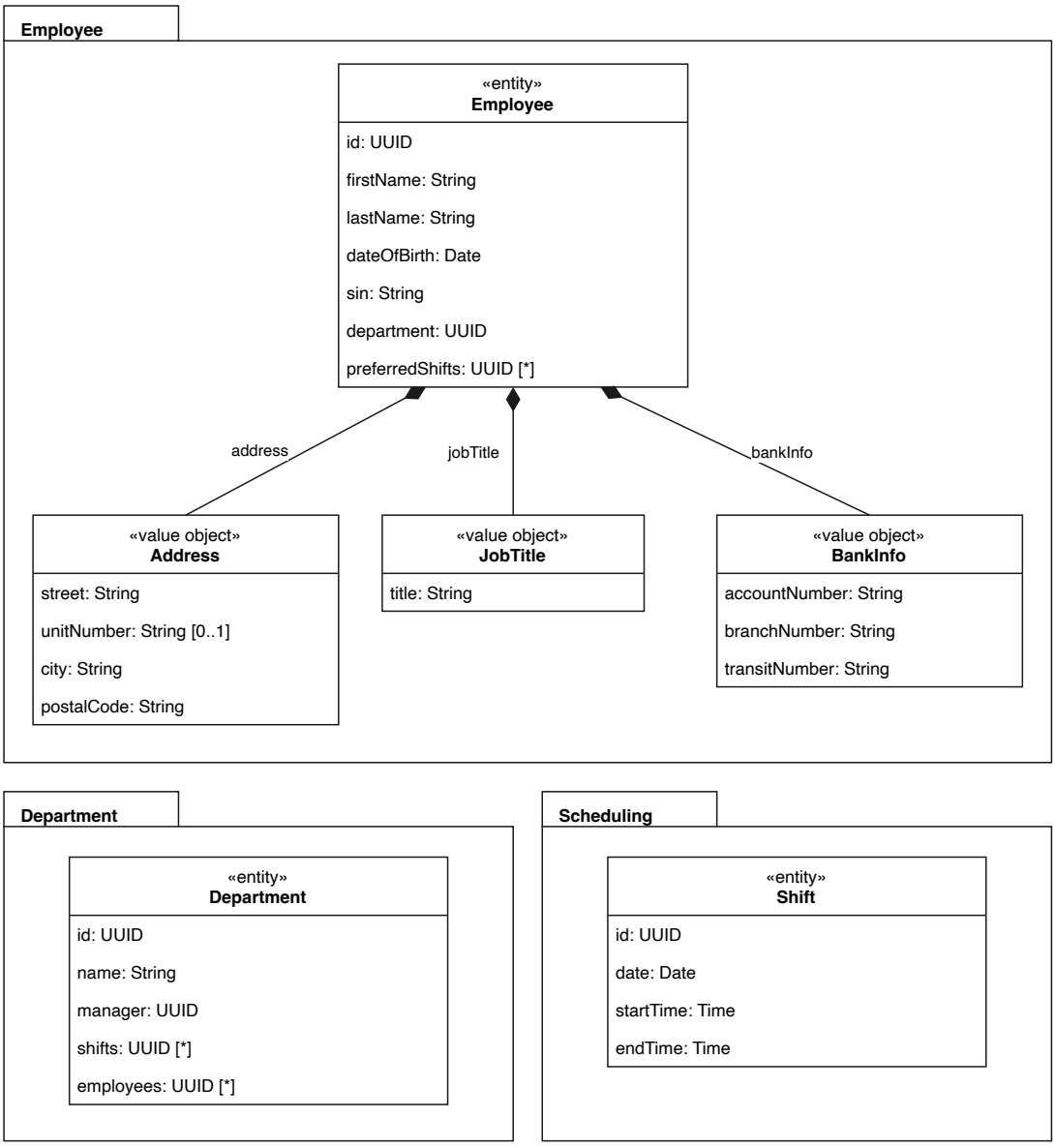


Figure 1: Domain model of AEMS for the two use cases

Employee, Department, and Shift are the roots of their aggregates. Aggregates refer to each other by id only:

- attribute `department` of Employee refers to a Department through its id
- attribute `preferredShifts` of Employee refers to a collection of Shifts through their ids
- attribute `shifts` of Department refers to a collection of Shifts through their ids
- attribute `employees` of Department refers to a collection of Employees through their ids

- attribute `manager` of Department refers to an Employee through its id

The registration information arrives in the data transfer object of Figure 2.

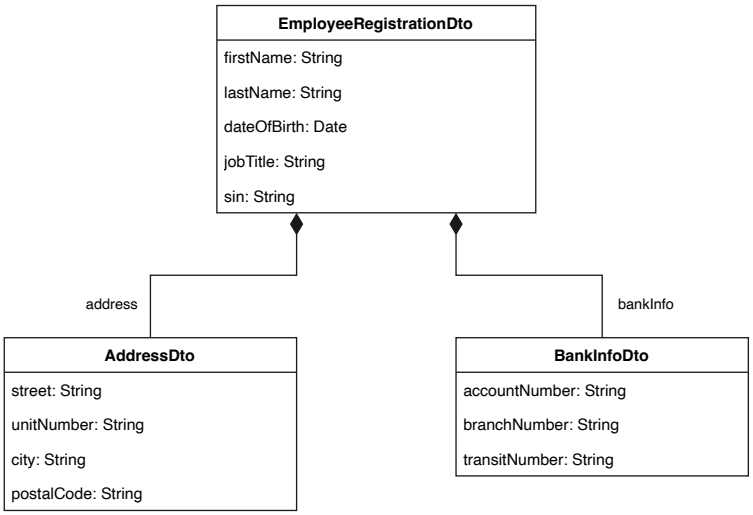


Figure 2: The registration DTO and its parts

Application command contracts

The following contracts were identified from the use cases *Register Department Employee* and *Select Work Shifts*.

Command: `registerEmployee(empInfo: EmployeeRegistrationDto)`	
Step	Step 3 of *Register Department Employee*
Preconditions	• the signed-in user is an Employee with job title Airport Manager • no Employee with the SIN given in `empInfo` exists
Postconditions	• a new Employee `emp` was created (instance creation) • the attributes of `emp` (first name, last name, date of birth, SIN) were initialized from `empInfo` (attribute modification) • the id of `emp` was set to a new employee id (attribute modification) • a new JobTitle `job` was created (instance creation) • the title of `job` was initialized from `empInfo.jobTitle` (attribute modification) • `job` was set as the job title of `emp` (association creation) • a new Address `addr` was created (instance creation) • the attributes of `addr` (street, city, postal code, optional unit number) were initialized from `empInfo.address` (attribute modification) • `addr` was set as the address of `emp` (association creation) • a new BankInfo `bank` was created (instance creation) • the attributes of `bank` (account number, branch number, transit number) were initialized from `empInfo.bankInfo` (attribute modification)

	• `bank` was set as the banking information of `emp` (association creation)

Command: `assignEmployeeToDepartment(empId: EmployeeId, deptId: DepartmentId)`	
Step	Step 6 of *Register Department Employee*
Preconditions	• an Employee with id `empId` exists • a Department with id `deptId` exists • the Employee with id `empId` is not yet assigned to a Department
Postconditions	• the Employee with id `empId` was added to the employees of the Department with id `deptId` (association creation) • the department of the Employee with id `empId` was set to the Department with id `deptId` (association creation)

Command: `selectShifts(empId: EmployeeId, shiftIds: ShiftId[])`	
Step	Step 3 of *Select Work Shifts*
Preconditions	• an Employee with id `empId` exists and is assigned to a Department • for each id in `shiftIds`, a Shift with that id exists, belongs to the Department of the Employee, and is upcoming (assumption: its date is later than today)
Postconditions	• each Shift with an id in `shiftIds` was added to the preferred shifts of the Employee with id `empId` (association creation)

Questions

Model each use case realization as UML sequence diagrams at two levels. At level 1, the application class of the use case interacts with the facades of the subdomains. At level 2, each facade interacts with the elements inside its subdomain: aggregate roots, factories, and repositories. Show where domain events are emitted.

Question 1 (65 marks)

Provide the use case realization of *Register Department Employee* at levels 1 and 2, based on the contracts of `registerEmployee` and `assignEmployeeToDepartment`.

Question 2 (30 marks)

Provide the use case realization of *Select Work Shifts* at levels 1 and 2, based on the contract of `selectShifts`.

Marking scheme

Question	Item	Marks
Question 1 (65 marks)	Level 1: scope, commands delegated to the right facades	20
	Level 2: postconditions of `registerEmployee` fulfilled	20

	Level 2: postconditions of `assignEmployeeToDepartment` fulfilled	15
	Level 2: scope, and every level 1 invocation expanded at level 2	10
Question 2 (30 marks)	Level 1: scope, command delegated to the right façade	10
	Level 2: postconditions of `selectShifts` fulfilled	10
	Level 2: scope, and every level 1 invocation expanded at level 2	10
Work distribution	Contribution of each member stated in a dedicated section	5
Total		100